Week 4 Model Evaluation Report

PM2.5 forecasting for 5 Cairo stations, horizons t+1 / t+2 / t+3 days | 2026-10-31 | protocol w4-v1

Status: Hold-out SCORED (see section 8). The model-selection weights (section 3) are a proposal; the picks are identical for any performance weight between 50% and 90%.

Summary

Final production models (one per horizon and task)

Horizon	Regression	CV RMSE	Classifier
t+1 (24h)	xgboost (pruned features)	14.87 +/- 5.18	XGBoost, alert at p >= 0.601
t+2 (48h)	lightgbm (pruned features)	16.91 +/- 6.14	XGBoost, alert at p >= 0.804
t+3 (72h)	lightgbm (pruned features)	17.53 +/- 6.69	XGBoost, alert at p >= 0.866

- **Tuning helped modestly.** Cross-validated RMSE fell by 3.1% / 3.9% / 3.3% for LightGBM and 2.5% / 3.0% / 2.1% for XGBoost (t+1 / t+2 / t+3), against the Week 3 defaults on the same folds.
- **LightGBM and XGBoost are statistically tied** for regression at every horizon (mean differences of 0.01-0.13 RMSE against a fold-to-fold std of 5-6.5, and no model wins 4 of 5 folds). The rubric therefore decided on speed and simplicity.
- **Accuracy degrades with the horizon:** CV RMSE 14.87 -> 16.91 -> 17.53 µg/m³ (+14% then +4%). The models beat the naive rule 'tomorrow = today' by only 4% at t+1, but by 16% at t+2 and 21% at t+3.
- **The health-risk alert does not reach the target of recall >= 0.90 at precision ~0.75.** With the precision floor of 0.75 the pooled out-of-fold recall is 0.42 / 0.17 / 0.16. At t+1 it is no better than the 'today's flag' rule at equal precision; at t+2 and t+3 it adds about 7 and 8 points of recall.
- **Errors are concentrated** in winter, at the Helwan station, and on the highest-PM2.5 days (Task 5). Per-district models and recursive forecasting did not help; averaging the two models gave a negligible gain.
- **Hold-out (Mar-Aug 2026, scored once):** RMSE 12.91 / 14.01 / 14.48 µg/m³. At t+1 this equals repeating today's value (12.84); at t+2 and t+3 it is 18% and 26% better. By season the hold-out error matches the CV, so there is no sign that the tuning over-fitted. At the fixed thresholds the alert has precision 0.74 / 0.75 / 0.40 but recall of only 0.13 / 0.02 / 0.02 (23 / 4 / 5 alerts for 130 risk days).

1. Data and evaluation protocol

- **Data:** 5 stations (downtown, helwan, new_cairo, october, shobra), daily. Tuning set 2023-09-08 to 2026-02-25 = 4,510 station-days (902 days x 5). Frozen hold-out 2026-03-01 to 2026-08-28 = 905 station-days (181 days x 5); 3 days between the two sets are dropped so no tuning label falls inside the hold-out.
- **Targets:** PM2.5 on day t+h (regression) and `is_health_risk` on day t+h (classification; risk = PM2.5 of about 55.4 µg/m³ or more, inferred from the data).
- **Cross-validation:** 5 rolling-origin folds of 90 test days (minimum 400 training days), pooled stations. Fold seasons by test period: winter (Dec-Mar), spring (Mar-May), summer (Jun-Aug), autumn (Sep-Nov), winter (Nov-Feb).
- **Verdict rule (used throughout):** a candidate counts as better only if its mean RMSE is lower AND it wins at least 4 of the 5 folds; otherwise it is a tie.
- **Hold-out discipline:** it was frozen in Task 1; nothing in Tasks 2-5 or in steps 1-4 of this task loads it. It is scored once, with the final models fixed beforehand.

2. Tuned versus default models

Regression: Week 3 default parameters vs tuned parameters, same folds and feature set (mean over 5 folds)

Model	Horizon	Default CV RMSE	Tuned CV RMSE	Change
lightgbm	t+1 (24h)	15.40	14.93	-3.1%
lightgbm	t+2 (48h)	17.60	16.91	-3.9%
lightgbm	t+3 (72h)	18.13	17.53	-3.3%
xgboost	t+1 (24h)	15.26	14.87	-2.5%
xgboost	t+2 (48h)	17.56	17.03	-3.0%
xgboost	t+3 (72h)	17.90	17.52	-2.1%

The defaults reproduce the Task 1 baseline exactly, so the comparison is like for like. The tuned scores are slightly optimistic because the parameters were chosen on the same folds (best of 60 trials).

Classification (XGBoost, pruned features): mean over 5 folds, recall at precision >= 0.75 and ROC AUC

Horizon	Default recall@P0.75	Tuned recall@P0.75	Default AUC	Tuned AUC
t+1 (24h)	0.24	0.36	0.789	0.806
t+2 (48h)	0.13	0.15	0.711	0.717
t+3 (72h)	0.04	0.14	0.663	0.645

A fold in which no threshold reaches precision 0.75 is scored as recall 0 and still enters the mean, which is why the fold-to-fold std of this metric is as large as the mean itself. Tuning optimised recall at the precision floor, not AUC: at t+3 the AUC is slightly lower after tuning (0.663 to 0.645) although recall at precision 0.75 is higher.

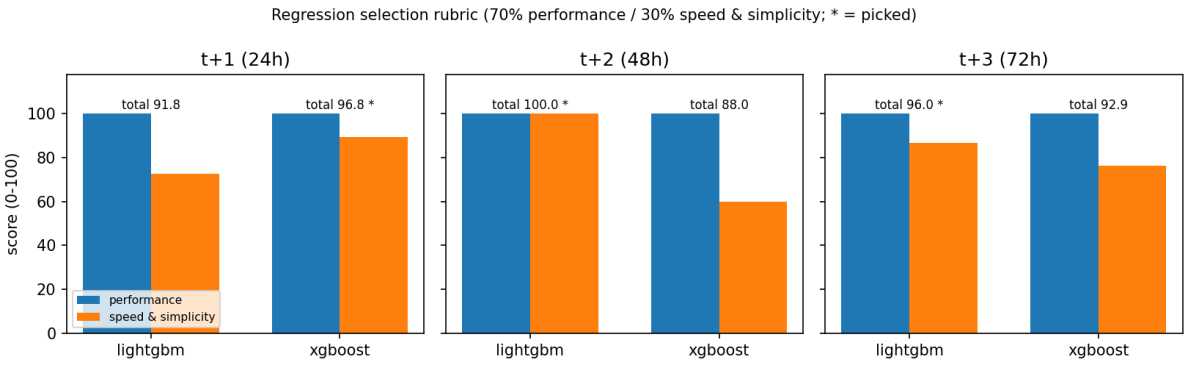
3. Model selection

Rubric (proposed weights: 70% performance, 30% speed and simplicity). The brief proposed 40% recall, 30% RMSE and 30% speed/simplicity. Recall only exists for classifiers and RMSE only for regressors, so no single model can be scored on both; the two tasks are ranked separately and the 40% + 30% are merged into one performance weight. The brief asks the team to agree the weights; the sensitivity table below shows the outcome does not depend on them within 50-90% performance weight.

- **Performance:** if one candidate has the lower mean RMSE AND wins at least 4 of 5 folds it scores 100 and the other 100 x RMSE ratio; otherwise it is a tie and both score 100.
- **Speed and simplicity:** average of three 0-100 ratios (best/value): inference time for 905 rows, model file size, number of input features.
- **Classification:** only one classifier family was tuned (XGBoost, Task 4; it tied with LightGBM there and was kept for simplicity), so there is nothing to rank; its threshold is fixed from Task 4.

Regression candidates and rubric scores

Horizon	Model	Features	CV RMSE	Fold wins	Latency ms	Size KB	Perf	Speed/simpl.	Total	Picked
t+1 (24h)	lightgbm	full (101)	14.93 +/- 5.41	2/5	39.2	2161	100	73	91.8	
t+1 (24h)	xgboost	pruned (79)	14.87 +/- 5.18	3/5	15.5	3169	100	89	96.8	YES
t+2 (48h)	lightgbm	pruned (79)	16.91 +/- 6.14	2/5	5.2	108	100	100	100.0	YES
t+2 (48h)	xgboost	pruned (79)	17.03 +/- 6.44	3/5	8.0	675	100	60	88.0	
t+3 (72h)	lightgbm	pruned (79)	17.53 +/- 6.69	3/5	14.1	378	100	87	96.0	YES
t+3 (72h)	xgboost	pruned (79)	17.52 +/- 6.51	2/5	8.5	1286	100	76	92.9	



Rubric scores per horizon

Sensitivity: the pick for performance weights from 50% to 100% (raw = ignoring the tie rule; 'tie (exact)' = identical scores)

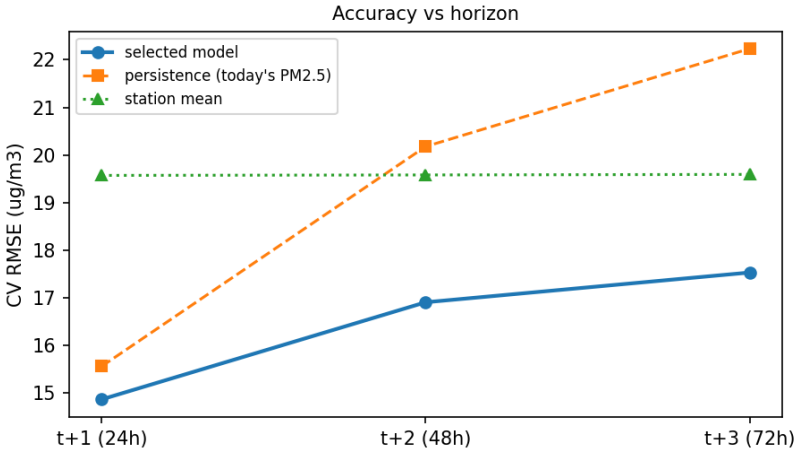
Horizon	Ranking	50%	60%	70%	80%	90%	100%
t+1 (24h)	raw	xgboost	xgboost	xgboost	xgboost	xgboost	xgboost
t+1 (24h)	tie rule	xgboost	xgboost	xgboost	xgboost	xgboost	tie (exact)
t+2 (48h)	raw	lightgbm	lightgbm	lightgbm	lightgbm	lightgbm	lightgbm
t+2 (48h)	tie rule	lightgbm	lightgbm	lightgbm	lightgbm	lightgbm	tie (exact)
t+3 (72h)	raw	lightgbm	lightgbm	lightgbm	lightgbm	lightgbm	xgboost
t+3 (72h)	tie rule	lightgbm	lightgbm	lightgbm	lightgbm	lightgbm	tie (exact)

The picks do not change for any weight between 50% and 90%. At 100% performance weight the candidates tie by construction. Because the two models are statistically tied on accuracy, the choice is driven by serving cost, not by a real accuracy difference. Mixing XGBoost (t+1) and LightGBM (t+2, t+3) is a consequence of that rule; picking one family for all horizons would cost at most about 0.13 RMSE in CV (XGBoost only, at t+2).

4. Accuracy by forecast horizon

CV RMSE of the selected regression models and two naive baselines (same folds)

Horizon	Selected model CV RMSE	vs previous horizon	Persistence (today's PM2.5)	Station mean	Skill vs persistence
t+1 (24h)	14.87	-	15.56	19.57	4.5%
t+2 (48h)	16.91	+13.7%	20.18	19.58	16.2%
t+3 (72h)	17.53	+3.7%	22.23	19.59	21.1%



Most of the loss happens between t+1 and t+2. At t+1 the best model is only about 4% better than repeating today's value, and in the summer fold repeating today's value is slightly better (8.05 vs 8.27); the models' added value is mainly at 48-72 h and in winter.

5. Health-risk alert: final thresholds

The alert fires when the predicted probability is at least the threshold. Thresholds come from Task 4 (the lowest-false-alarm point with precision >= 0.75 on pooled out-of-fold predictions) and are **not re-tuned on the hold-out**. The confusion matrices below were rebuilt in this task from the same folds; they reproduce Task 4's recall and precision to within 0.03 (XGBoost is not bit-for-bit reproducible across library versions).

Confusion matrices at the chosen thresholds (pooled out-of-fold, 2,250 station-days per horizon; this AUC is pooled, Task 4 reports the per-fold mean 0.806 / 0.717 / 0.645)

Horizon	Threshold	Imbalance	TP	FP	FN	TN	Precision	Recall	AUC
t+1 (24h)	0.601	none	254	78	351	1567	0.765	0.420	0.864
t+2 (48h)	0.804	class_weight	102	34	502	1612	0.750	0.169	0.811
t+3 (72h)	0.866	class_weight	96	36	506	1612	0.727	0.159	0.763

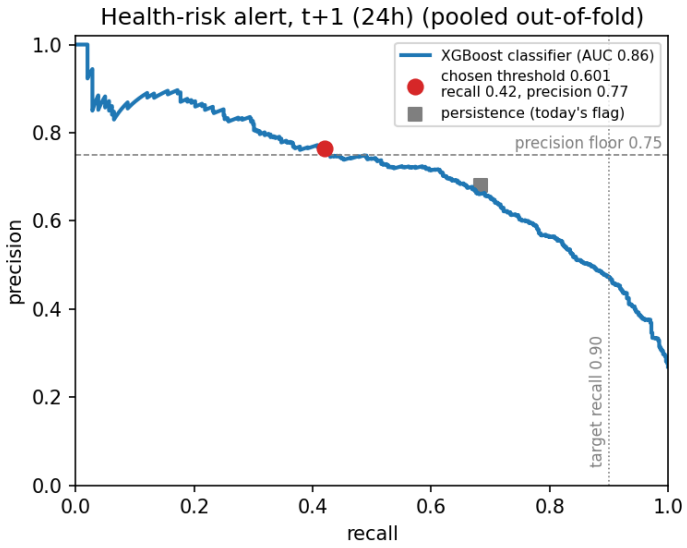
In plain language: what happens per 100 station-days

Horizon	Risk days	Alerts issued	Correct alerts	False alarms	Risk days missed
t+1 (24h)	26.9	14.8	11.3	3.5	15.6
t+2 (48h)	26.8	6.0	4.5	1.5	22.3
t+3 (72h)	26.8	5.9	4.3	1.6	22.5

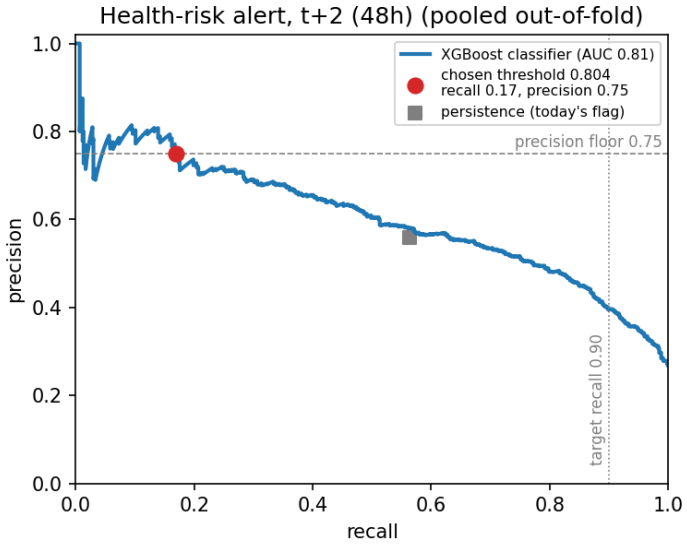
- **t+1:** of every 100 station-days about 27 are risk days; the alert fires on 15, of which 3.5 are false alarms (about 1 in 4 alerts), and about 16 risk days are missed (recall 42%).
- **t+2 and t+3:** the alert is very conservative (only about 6 alerts per 100 days) and misses about 22 risk days out of the 27. It is correct when it fires, but it is silent for most episodes.
- **Target not met:** the brief asked for recall >= 0.90 at precision around 0.75. In the cross-validation this is not reachable with these models at any horizon; the numbers above are the honest operating point at precision 0.75.

Naive comparison: 'tomorrow is a risk day if today is' vs the model at the same precision (pooled out-of-fold)

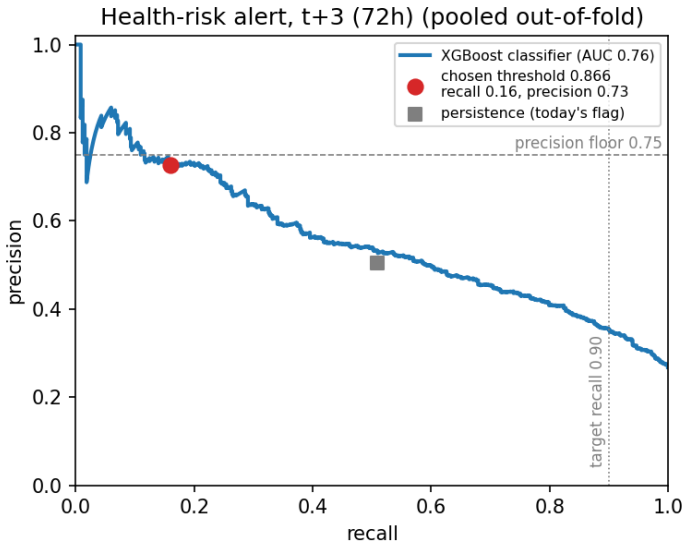
Horizon	'Today's flag' precision	'Today's flag' recall	Model recall at that precision	Difference
t+1 (24h)	0.681	0.684	0.658	-2.6 pts
t+2 (48h)	0.559	0.563	0.631	+6.8 pts
t+3 (72h)	0.503	0.508	0.588	+8.0 pts



Precision-recall curve, t+1 (24h)



Precision-recall curve, t+2 (48h)



Precision-recall curve, t+3 (72h)

Per fold at the fixed thresholds: the alert almost never fires outside winter (a '-' in precision means no alert fired)

Horizon	Fold	Season	Risk days	Alerts	TP	FP	FN	Precision	Recall
t+1 (24h)	1	winter	180	102	84	18	96	0.82	0.47
t+1 (24h)	2	khamsin	113	50	32	18	81	0.64	0.28
t+1 (24h)	3	other	32	1	0	1	32	0.0	0.0
t+1 (24h)	4	other	49	17	7	10	42	0.41	0.14
t+1 (24h)	5	winter	231	162	131	31	100	0.81	0.57
t+2 (48h)	1	winter	179	68	57	11	122	0.84	0.32
t+2 (48h)	2	khamsin	113	1	1	0	112	1.0	0.01
t+2 (48h)	3	other	32	0	0	0	32	-	0.0
t+2 (48h)	4	other	52	0	0	0	52	-	0.0
t+2 (48h)	5	winter	228	67	44	23	184	0.66	0.19
t+3 (72h)	1	winter	179	62	45	17	134	0.73	0.25
t+3 (72h)	2	khamsin	111	3	1	2	110	0.33	0.01
t+3 (72h)	3	other	32	0	0	0	32	-	0.0
t+3 (72h)	4	other	53	3	1	2	52	0.33	0.02
t+3 (72h)	5	winter	227	64	49	15	178	0.77	0.22

Consequence for the hold-out (March-August, no winter): very few risk days and very few alerts are expected, so hold-out recall and precision rest on a small number of events and should be read with their confidence intervals. This is what happened (section 8).

6. Where the models fail (error analysis from Task 5)

Tuned LightGBM CV RMSE by station

Station	t+1 (24h)	t+2 (48h)	t+3 (72h)
downtown	15.39	17.09	17.74
helwan	20.01	22.62	23.28
new_cairo	13.29	15.1	15.89
october	13.24	15.22	16.12
shobra	16.48	18.84	19.8

Tuned LightGBM CV RMSE by season (khamsin = Mar-May, other = Jun-Nov)

Season	t+1 (24h)	t+2 (48h)	t+3 (72h)
khamsin	14.23	16.17	16.57
other	9.77	11.06	11.3
winter	21.02	23.81	25.06

- Helwan is the hardest district: tuned LightGBM RMSE 20.0 / 22.6 / 23.3 (t+1 / t+2 / t+3), 33-34% above the overall RMSE, and it is the only district that is clearly under-predicted (bias -3.3 to -5.5 µg/m³). New Cairo and October are about 10% below overall, Shobra 10-13% above, Downtown about average. Helwan's mean PM2.5 (53) is similar to Downtown (55)

and Shobra (53), so its larger error is not just a higher level.

- Winter is the hard season: RMSE 21.0 / 23.8 / 25.1 vs 9.8 / 11.1 / 11.3 for Jun-Nov and 14.2 / 16.2 / 16.6 for Khamsin (Mar-May). In winter the models under-predict (bias -2.4 at t+1, -4.7 at t+3); in Khamsin and the rest of the year they over-predict slightly (+1.5 to +2.4).
- Peaks are badly under-predicted: on the top 5% PM2.5 days (>= 88.8 µg/m³) the models are low by 37-47% on average, 98-100% of those days are under-predicted, and they cause 43-46% of all squared error. This fits the low alert recall found in Task 4.
- October does not stand out as a failure: the October station is among the best predicted (-11% vs overall at t+1), and calendar October has RMSE about 10-12 with a positive bias (+4.9 to +6.5), i.e. slight over-prediction.
- Per-district models do not help: 1 of 6 model/horizon cells improved (XGBoost t+2, -0.9%, 5/5 folds), 3 were worse (up to +2.1%) and 2 tied. Helwan, the case this was meant to fix, did not improve in any of its 6 cells. Keep pooled models.
- The LightGBM + XGBoost average gives a very small gain: tuned -0.18% (t+1), +0.04% (t+2, tie), -0.43% (t+3), i.e. 0.03-0.08 RMSE, with 4/5 fold wins at t+1 and t+3. That is tiny next to the fold-to-fold std of 5-6.5 and costs two models per horizon. The 0.5 / 0.5 weight was fixed in advance; the weight sweep is informational only.
- Recursive multi-horizon does not help: 0 of 4 cells improved, 3 were worse (+0.6% to +1.2%) and 1 tied. Keep direct forecasting (one model per horizon).

Implication for the final models: keep pooled stations and direct forecasting (one model per horizon). Averaging LightGBM and XGBoost gains at most 0.08 RMSE and would double the models to serve, so it is not used.

7. Why deep-learning and time-series models were not tuned

This section relies on the Week 3 comparison reported in the team brief (the Week 3 workbook itself was not re-run or re-inspected here, and its folds differ from the Week 4 folds, so its absolute numbers are not comparable with the rest of this report).

- In Week 3, gradient boosting had the lowest RMSE at every horizon, then Random Forest, then GRU/LSTM, then Prophet/SARIMA. At t+1 LightGBM was reported about 33% better than SARIMA (10.2 vs 15.2) and about 23% better than GRU (10.2 vs 13.2).
- The signal sits in the weather, satellite and spatial features, and the lag features already give tree models the short-term memory an LSTM would add, so the expected gain from tuning a neural network was low and its cost high.
- Week 4 adds a sanity check: in the same cross-validation the simplest forecasters are clearly worse than the tuned boosters at 48-72 h (persistence RMSE 20.2 / 22.2 vs 16.9 / 17.5), while at 24 h persistence is already close (15.6 vs 14.9).
- Random Forest had the best classification AUC at t+3 in Week 3 (0.710 vs 0.645 for the tuned XGBoost here); it was kept only as a reference because the plan was to tune LightGBM and XGBoost.

8. Hold-out results

Scored once on the frozen hold-out (2026-03-01 to 2026-08-28) with the models fixed in section 3 and the Task 4 thresholds. Confidence intervals resample whole weeks.

Regression on the hold-out

Horizon	Model	n	RMSE	MAE	R2	Bias	Persistence RMSE	Station-mean RMSE
t+1 (24h)	t1_xgboost	905	12.91 [10.29, 15.64]	8.69	0.32	-0.40	12.84	16.95
t+2 (48h)	t2_lightgbm	905	14.01 [11.17, 17.24]	9.80	0.20	+0.06	17.16	16.99
t+3 (72h)	t3_lightgbm	905	14.48 [11.61, 17.93]	10.26	0.14	+0.64	19.48	16.99

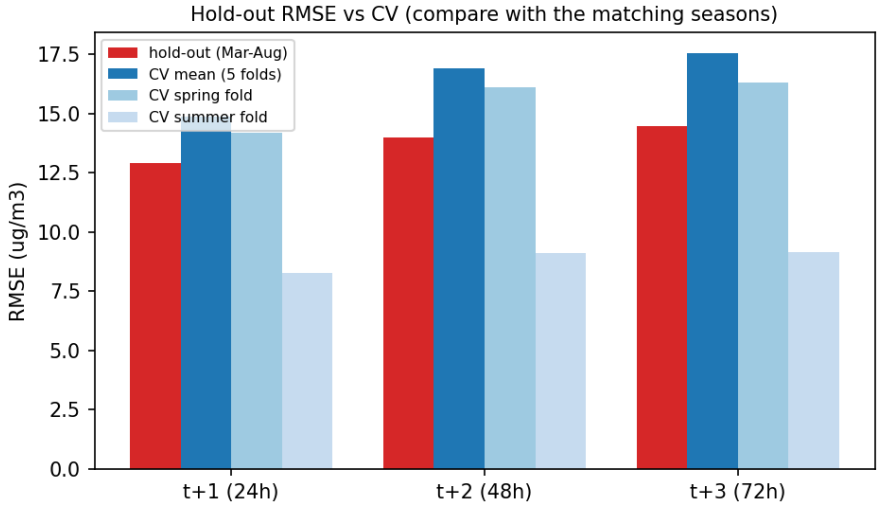
Classification on the hold-out at the fixed thresholds

Horizon	Threshold	Risk days	Alerts	TP	FP	FN	Precision	Recall [95% CI]	AUC	Today's-flag recall
t+1 (24h)	0.601	130 of 905	23	17	6	113	0.739	0.131 [0.04, 0.20]	0.804	0.554
t+2 (48h)	0.804	130 of 905	4	3	1	127	0.750	0.023 [0.00, 0.08]	0.753	0.346

Horizon	Threshold	Risk days	Alerts	TP	FP	FN	Precision	Recall [95% CI]	AUC	Today's-flag recall
t+3 (72h)	0.866	130 of 905	5	2	3	128	0.400	0.015 [0.00, 0.04]	0.713	0.238

Hold-out vs cross-validation (compare with the spring and summer folds, not only the CV mean)

Task	Horizon	Metric	Hold-out	CV mean (5 folds)	CV spring fold	CV summer fold	Persistence on hold-out
regression	1	RMSE	12.911	14.87	14.188	8.268	12.836
regression	2	RMSE	14.011	16.909	16.125	9.125	17.164
regression	3	RMSE	14.481	17.533	16.321	9.173	19.476
classification	1	recall at fixed threshold	0.131	0.292	0.283	0.0	0.554
classification	1	precision at fixed threshold	0.739	0.537	0.64	0.0	0.554
classification	2	recall at fixed threshold	0.023	0.104	0.009	0.0	0.346
classification	2	precision at fixed threshold	0.75	0.832	1.0	-	0.346
classification	3	recall at fixed threshold	0.015	0.099	0.009	0.0	0.238
classification	3	precision at fixed threshold	0.4	0.54	0.333	-	0.238



Hold-out RMSE by station

Station	t+1 (24h)	t+2 (48h)	t+3 (72h)
downtown	10.79	11.36	12.4
helwan	15.76	16.92	17.26
new_cairo	9.16	10.03	10.72
october	15.41	17.04	17.11
shobra	12.13	13.23	13.75

- **No sign of over-fitting.** By season the hold-out RMSE matches the tuned CV (March-May 15.2 / 16.4 / 17.1 vs CV 14.2 / 16.2 / 16.6; June-August 10.0 / 11.0 / 11.2 vs CV 9.8 / 11.1 / 11.3; largest gap 0.9 µg/m³). The overall hold-out RMSE is lower than the CV mean only because the hold-out contains no winter.
- **At t+1 the regression is not better than repeating today's value** (RMSE 12.91 vs 12.84); at t+2 and t+3 it is 18% and 26% better. R2 is low (0.32 / 0.20 / 0.14) because PM2.5 varies much less in the warm months, so the same absolute error explains less of the variance. Bias is small (-0.4 / +0.1 / +0.6 µg/m³).
- **Stations:** new_cairo is predicted best and helwan worst at t+1. The October station is clearly worse here (15.4 at t+1) than in the CV (13.2), where it was among the best; Helwan stays hard (15.8 vs 20.0 in CV).
- **Alert:** at the fixed thresholds precision is 0.74 at t+1 (target 0.75) but the alert fires only 23 / 4 / 5 times in 905 station-days with 130 risk days, so recall is 0.13 / 0.02 / 0.02 (95% intervals above). All alerts fall in March-May; June-August has 35 risk days and no alert at any horizon. At t+2 and t+3 the precision rests on 4 and 5 alerts and is not meaningful.

- **The ranking quality of the classifier holds up** (hold-out AUC 0.80 / 0.75 / 0.71 vs Task 4's CV means 0.81 / 0.72 / 0.65): the models still separate risk from non-risk days, but the thresholds, set for a precision of 0.75 on winter-dominated folds, are too high for the warm months. A 'today's flag' rule catches 55% of risk days at t+1 but with precision 0.55. The thresholds were NOT changed after seeing the hold-out; any recalibration must be done on CV data and would need new, untouched data to be confirmed.
- **For the Week 5 deployment:** the regression forecasts are useful at 48-72 h and no better than persistence at 24 h in the warm season; the risk alert, as calibrated, is not a dependable early warning outside winter and should be presented with its recall.

Scored once on 2026-10-31T17:14:36; the lock file prevents a second scoring.

9. Production models and serialization

Serialized models (models/*.pkl and .onnx, model_cards.json has features, parameters, threshold and library versions)

File	Features	Trained on	pkl reload diff	ONNX	SHA-256
pm25_reg_t1_xgboost	79	4510 rows (2023-09-08 to 2026-02-25)	0e+00	OK, max diff 2.1e-04	f923bd5f3e
pm25_clf_t1_xgboost	79	4510 rows (2023-09-08 to 2026-02-25)	0e+00	OK, max diff 4.2e-07	291360cb66
pm25_reg_t2_lightgbm	79	4510 rows (2023-09-08 to 2026-02-25)	0e+00	OK, max diff 4.4e-05	8201c53911
pm25_clf_t2_xgboost	79	4510 rows (2023-09-08 to 2026-02-25)	0e+00	OK, max diff 8.9e-08	27b9684b77
pm25_reg_t3_lightgbm	79	4510 rows (2023-09-08 to 2026-02-25)	0e+00	OK, max diff 7.6e-02	def420e28d
pm25_clf_t3_xgboost	79	4510 rows (2023-09-08 to 2026-02-25)	0e+00	OK, max diff 8.9e-08	3db061104e

- Each model was retrained on the whole tuning set (all rows with a label) with the tuned parameters, then reloaded from disk: predictions are identical to the in-memory model (difference 0).
- ONNX copies reproduce the sklearn models to within float32 rounding (largest difference 0.08 µg/m³, on the t+3 LightGBM model, against an RMSE of about 17).
- MLflow: `task6_04_register_mlflow.py` logs the six models to experiment PM25_week4_tuning and registers them as pm25_reg_t1..t3 and pm25_clf_t1..t3. It was tested on a scratch database (every model reloaded from the registry with difference 0). **It still has to be run once on the team's real MLflow database.**
- Inference use: regression `model.predict(X[features])`; classification `model.predict_proba(X[features])[:, 1] >= alert_threshold` (threshold in model_cards.json).

10. Limitations and open points

- **Optimism:** tuned scores and thresholds were chosen on the same CV folds; expect slightly worse numbers on new data.
- **Classifier:** recall at precision 0.75 is low (0.42 / 0.17 / 0.16 pooled) and unstable across folds (many folds have no valid threshold). The Week 5 deployment should not present the 48-72 h alert as reliable.
- **Small tied differences:** LightGBM vs XGBoost, full vs pruned features and per-district vs pooled are all within noise except where stated; the selection reflects serving cost.
- **Reproducibility:** XGBoost results differ by up to about 0.04 RMSE (0.25%) between runs on different library versions; the Task 2 report's printed parameter block does not reproduce its scores (the Optuna database and the JSON file do), so the Optuna database was used.
- **Only one classifier family** (XGBoost) was tuned; a LightGBM classifier result was not available in the Task 4 hand-off.
- **Week 3 comparisons** quoted in section 7 come from the team brief and were not re-verified.
- **Hold-out:** it contains no winter and its 130 risk days sit in March-May, so it can confirm the regression (seasonal errors match the CV) but says little about the winter performance of the alert; intervals are wide because days within a week are strongly correlated.
- **Weights** of the selection rubric were proposed by the author; the outcome is insensitive to them (50-90% performance weight) but they should be confirmed with the team.

11. Reproducing this report

- `python task6_01_evidence.py` (CV evidence, baselines, timing, confusion matrices; a few minutes)

- `python task6_02_select.py [--w-perf 0.7]` (rubric and sensitivity)
- `python task6_03_train_save.py` (train, serialize, reload checks)
- `python task6_04_register_mlflow.py` (MLflow registry, run on the team database)
- `python task6_05_score_holdout.py --confirm TASK6_FINAL_SCORING` (ONE time)
- `python task6_06_build_report.py` (this report)

Appendix: selected hyperparameters

Model	Parameters
reg_t1_xgboost	max_depth=7, learning_rate=0.01146, n_estimators=650, min_child_weight=3.915, subsample=0.5192, colsample_bytree=0.4373, gamma=8.515e-06, reg_lambda=0.008233
clf_t1_xgboost	max_depth=7, learning_rate=0.01476, n_estimators=250, min_child_weight=8.387, subsample=0.5879, colsample_bytree=0.8938, gamma=0.0004212, reg_lambda=0.004964
reg_t2_lightgbm	num_leaves=10, learning_rate=0.05288, n_estimators=100, min_child_samples=59, subsample=0.6151, colsample_bytree=0.828, reg_alpha=5.471e-07, reg_lambda=1.992e-06, subsample_freq=1
clf_t2_xgboost	max_depth=7, learning_rate=0.02441, n_estimators=100, min_child_weight=14.79, subsample=0.787, colsample_bytree=0.7772, gamma=0.01425, reg_lambda=0.09831
reg_t3_lightgbm	num_leaves=13, learning_rate=0.01555, n_estimators=300, min_child_samples=24, subsample=0.5837, colsample_bytree=0.7433, reg_alpha=2.163e-07, reg_lambda=0.1305, subsample_freq=1
clf_t3_xgboost	max_depth=5, learning_rate=0.0705, n_estimators=350, min_child_weight=12.04, subsample=0.5894, colsample_bytree=0.9862, gamma=0.0009087, reg_lambda=0.4675